

Atmospheric-neutrino truth report

What this sample represents

Each entry is a generated neutrino-argon interaction that was handed to Geant4. The incoming-neutrino distributions are therefore interaction distributions shaped by the atmospheric flux, the neutrino and antineutrino cross sections, the argon target, generator sampling, and any configured weights. They are not a direct plot of the unmodified atmospheric flux.

Truth extraction

Exactly one PDG 12, -12, 14, or -14 incoming neutrino was found in every one of the 10,000 events. Energy and direction come from `generator_initial_particle_energy` and the initial `px`, `py`, `pz` branches. Final-state multiplicity and charged-current-like classifications use the generator final-state PDG list, before Geant4 creates transport secondaries.

Flavor accounting

The interaction counts are $\nu_e=2,972$ (29.7%), $\bar{\nu}_e=646$ (6.5%), $\nu_\mu=4,946$ (49.5%), $\bar{\nu}_\mu=1,436$ (14.4%). The unweighted neutrino-to-antineutrino interaction ratio is 3.80. A neutrino excess is physically reasonable because the atmospheric flux and interaction cross sections both differ between neutrinos and antineutrinos.

Interpretation caution

Do not compare these raw flavor bars directly with a published flux table unless the generator exposure, solid-angle integration, cross-section sampling, oscillations, and event weights are treated consistently. Their strongest immediate use is bookkeeping and regression validation of this exact production chain.

Energy and direction interpretation

Incoming energy

The overall median incoming energy is 735 MeV, the mean is 1301 MeV, and the central 90% interval is 193 to 4648 MeV. Per-flavor medians are $\nu_e=652$ MeV, $\text{anti-}\nu_e=723$ MeV, $\nu_\mu=760$ MeV, $\text{anti-}\nu_\mu=881$ MeV; per-flavor means are $\nu_e=1133$ MeV, $\text{anti-}\nu_e=1230$ MeV, $\nu_\mu=1344$ MeV, $\text{anti-}\nu_\mu=1530$ MeV. The mean is larger than the median when the high-energy tail is influential; the median is therefore the steadier typical-event metric, while the mean describes the sample's total energy budget. Differences among flavors can arise from flavor-dependent atmospheric fluxes and charged- and neutral-current interaction probabilities.

Flavor fraction versus energy

The stacked fraction panel reveals whether flavor composition changes across energy without allowing the steep total spectrum to dominate the view. Smooth evolution is physically plausible; isolated jumps in the sparsely populated highest-energy bins should be interpreted as finite-sample fluctuations.

Detector-coordinate direction

The $\cos(\theta_z)$ plot uses the simulation z axis. It becomes a conventional atmospheric zenith plot only after confirming the flux-to-detector coordinate mapping and sign convention. Atmospheric fluxes can have real zenith dependence, and oscillations can strongly reshape flavor and zenith distributions, so exact flatness is not generally required. Per-flavor median/mean $\cos(\theta_z)$ values are $\nu_e=-0.05/-0.05$, $\text{anti-}\nu_e=-0.06/-0.07$, $\nu_\mu=-0.07/-0.06$, $\text{anti-}\nu_\mu=-0.05/-0.05$.

Azimuth and energy-angle structure

Azimuth should be approximately uniform when the generation has no detector-orientation weighting. The energy-angle density and median-energy profile expose directional energy biases that separate one-dimensional plots can hide. Any strong periodic azimuth structure would warrant checking the flux rotation and momentum-coordinate transformation.

Interaction outcome assessment

Charged-current-like classification

The expected charged lepton appears in 6,663 of 10,000 events (66.6%). Per-flavor fractions are $\nu_e=70.1\%$, $\text{anti-}\nu_e=57.4\%$, $\nu_\mu=68.5\%$, $\text{anti-}\nu_\mu=57.2\%$. The remainder are predominantly neutral-current-like under this PDG-based definition. This is a truth-topology inference, not a direct readout of GENIE's interaction-mode code.

Final-state complexity

The median generator final-state multiplicity is 3, the mean is 5.65, and the central 90% spans 2 to 18 particles. Its correlation with $\log_{10}$ incoming energy is 0.11. A positive trend is physically expected because higher-energy interactions have more phase space for resonance production, deep-inelastic scattering, and larger hadronic systems.

Overall assessment

The flavor accounting, broad energy spectra, populated all-sky directions, approximately uniform azimuth, charged-lepton sign logic, and increasing final-state complexity provide complementary generator sanity checks. None alone establishes an absolute atmospheric event rate, but together they can reveal unit errors, missing flavors, direction-convention mistakes, or broken generator-to-Geant4 truth handoff.

Recommended quantitative validation

Preserve event weights and compare the incoming energy-flavor-zenith distribution against the exact Honda table and generation configuration. Then compare charged-current and neutral-current fractions and final-state observables under GENIE tune and cross-section systematic variations. Apply oscillation probabilities before comparing with an underground detector's expected atmospheric-neutrino data.